Quiz Week 11

Factor groups

$$
\begin{tikzcd}
G \arrow[r, "f"] \arrow[d, "\pi"'] & H \\
G/\ker(f) \arrow[ur, dashed, "\tilde{f}"'] &
\end{tikzcd}
$$

Normal subgroups. Factor groups

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1. Select which of the following subgroups is normal in $G = GL(2,\mathbb{R})$.

   (a) $H = \left\{ \begin{pmatrix} a & b \\ 0 & d \end{pmatrix} : a, b, d \in \mathbb{R},\ ad \neq 0 \right\}$;

   (b) $H = \{A \in GL(2,\mathbb{R}) : \det(A) \in K\}$, for K a subgroup of $\mathbb{R}\backslash\{0\}$.

Solution: (a) H is not normal. In fact, let $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \in H$ and $X = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} \in GL(2,\mathbb{R})$. Then $X^{-1} = \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix}$ and $XAX^{-1} = \begin{pmatrix} 0 & 1 \\ -1 & 2 \end{pmatrix} \notin H$. Therefore, H is not normal in $GL(2,\mathbb{R})$.

(b) H is normal in G. Let us first show that H is a subgroup of G. Notice that H contains the identity since $1 \in K$. If A and B are in H, then $\det(A), \det(B) \in K$. Applying that K is a subgroup of $\mathbb{R}\backslash\{0\}$, we have that $\det(A)\det(B) \in K$, that is, $\det(AB) \in K$ and so $AB \in H$. Lastly, $\det(A^{-1}) = \det(A)^{-1} \in K$ and so $A^{-1} \in K$.

To show that H is normal in $GL(2,\mathbb{R})$, we will use the *Normality Test*. Let $A \in H$ and $X \in GL(2,\mathbb{R})$, then $\det(A) \in K$ and

$$\det(XAX^{-1}) = \det(X)\det(A)\det(X^{-1}) = \det(X)\det(X)^{-1}\det(A) \in K,$$

which implies that $XAX^{-1} \in H$. Thus $H \lhd GL(2,\mathbb{R})$.